

Tarot Wheel: Designing Embodied Ritual Through Gesture-Controlled Actuation

CMSC 20380 1 Actuated User Interfaces and Technology

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5.26.2026

1. Overall Idea and Concept of the Project

My project, Fortune Carousel, is an actuated tarot interaction device inspired by the visual and emotional experience of a carousel. A carousel is circular, rhythmic, and slightly dreamlike: objects pass in front of the viewer one by one, and the moment of choosing depends on timing, attention, and attraction. I connected this image with tarot reading, an ancient ritual in which people usually select a card from a spread placed on the table. In a traditional tarot reading, all the card backs may look the same, but the participant is asked to choose the one they feel drawn to. I was interested in how this moment of choosing could become more dynamic, embodied, and magical.

Instead of presenting tarot cards flat on a table, my project places the cards upright on a rotating platform. As the cards move in front of the user, selection becomes less about looking at a static spread and more about sensing timing, connection, and attraction. The user does not simply pick a card that is already lying still; they encounter the cards as moving objects. The chosen card appears through a specific moment of alignment between the user's gesture, the motion of the turntable, and the timing of the stop.

The interaction is designed to feel like gently “pushing” or “sliding” a carousel of cards through the air. When ultrasonic sensors detect a hand movement within 30 cm, a stepper motor drives the turntable clockwise or counterclockwise. The user can hover, wave, and guide the direction of rotation without touching the device. This non-contact control adds to the magical quality of the experience: the user’s hand appears to influence the movement of the cards at a distance, as if casting a small spell.

The final selection gesture uses both hands. When both hands move very close to the sensors, within approximately 10 cm, the system interprets this as adding force or intention to stop the turntable. The platform immediately stops, and a short vibration reminds the user to take the selected card. In this way, the system turns a simple technical mechanism into a ritual sequence: sensing, rotating, feeling attraction, stopping, and receiving the card.

For the prototype, I simplified the tarot deck into seven representative cards. This was not meant to reproduce a complete tarot system, but to create a compact “everyday tarot” suitable for a short public interaction. The seven-card set keeps the symbolic feeling of tarot while making the physical rotating mechanism feasible and readable in an exhibition setting.

2. Motivation and Concept for Building the Interactive Experience

The motivation for this project came from my interest in tarot as a physical and emotional experience. Tarot reading is not only about the final card or its written meaning. The experience also comes from waiting, touching, choosing, and imagining that the object carries hidden meaning. I wanted to preserve this feeling, but make the selection process more dynamic through actuation.

In many digital fortune-telling systems, the user clicks a button and immediately receives a result. I wanted to avoid this direct and efficient interaction model. Instead, I wanted to create a slower and more atmospheric experience. The rotating platform delays the final result and makes the user watch, wait, and sense the right moment. The motion becomes part of the meaning of the reading.

This motivation connects to Klefeker et al.'s discussion of enchantment in HCI. Their work on ASMR-inspired design argues that slow, intimate, embodied, and sensory experiences can help people become “enchanted with the mundane” . My project similarly tries to make an ordinary tabletop object feel mysterious and affective. The goal is not to make tarot faster or more efficient, but to turn the moment of choosing into a felt experience.

The project also relates to tangible and multimodal emotional interfaces. In Bondi, Xiong et al. designed a tangible and multimodal artifact that uses touch, sound, light, warmth, and movement to support emotional and symbolic connection . Although my project is not about bereavement, it similarly treats a physical object as a carrier of emotion, memory, and symbolic interpretation.

The tarot turntable is not only a machine for selecting a card; it is also a ritual object that invites the user to project meaning onto its movement.

3. Grounding in the Theme: “Actuated Interactive Everyday Matter — Designing Interaction via Form, Motion, and Touch”

This project is grounded in the course theme because it uses form, motion, and touch as the main materials of interaction.

First, form is central to the project. The circular structure of the turntable recalls both a carousel and a wheel of fate. The cards are placed upright around the platform, so the object does not look like a normal flat tarot spread. Instead, the vertical arrangement creates a small stage where cards appear, disappear, and return in cycles. This physical form supports the theme of fate, timing, and ritual.

Second, motion is the primary interaction output. The stepper motor rotates the turntable according to hand gestures detected by the sensors. The rotating motion is not just functional; it is expressive. It creates anticipation and gives the object a sense of agency. Zuckerman et al. describe how actuated interfaces can communicate through physical movement and how shared-control actuated interfaces raise questions of coordination, timing, and human autonomy . In my project, the user and the object share control over the moment of selection: the user guides the

rotation through gesture, while the system produces the physical motion, interprets distance thresholds, and determines when the stopping and vibration feedback happen.

This shared control is especially important to the tarot experience. The user can influence the direction and timing of rotation, but the final moment of stopping still feels partially outside their direct control. This creates an intentional ambiguity between “I am controlling it” and “the object is choosing for me.” Rather than treating this ambiguity as a technical weakness, I use it as part of the ritual. The slight uncertainty supports the feeling of fate, chance, and timing.

Third, touch appears through vibration feedback. The user does not directly press a button, but when the selection gesture is detected, the vibration motor gives a short tactile signal. This creates a bridge between non-contact sensing and bodily feedback. Uhl et al. argue that tangible and haptic interaction can make abstract experiences more intuitive by connecting them to physical action and sensation . In my project, the abstract act of “choosing fate” becomes something visible through rotation and felt through vibration.

Together, these elements turn an everyday object into actuated interactive matter. The object senses, moves, stops, and vibrates. It is not only a display for tarot cards; it becomes a responsive physical interface that uses form, motion, and touch to create an embodied ritual.

4. Technical Detail of the System

The system is built with an Arduino Uno, two HC-SR04 ultrasonic distance sensors, a 28BYJ-48 stepper motor, a ULN2003 stepper motor driver, a coin vibration motor, a breadboard, jumper wires, and a handmade rotating card platform. The electronic system reads distance values from the ultrasonic sensors and uses those values to control the stepper motor and vibration feedback.

The two ultrasonic sensors are placed on opposite sides of the turntable. Each sensor creates an invisible interaction zone. When a user's hand enters the sensing range within approximately 30 cm, the system interprets this as a directional gesture. One sensor controls clockwise rotation, and the other controls counterclockwise rotation. This allows users to "push" the card carousel in either direction without touching it.

When both hands move close to the two sensors at the same time, within approximately 10 cm, the system interprets the gesture as a selection. The turntable immediately stops, and the vibration motor gives two short pulses, each lasting 0.3 seconds. After this feedback, the system pauses briefly, so the stop feels intentional rather than accidental. The user can then take the card positioned at the selection point.

The stepper motor is used because the turntable needs continuous circular movement. A standard angle servo was not suitable for the interaction because it tended to rotate between fixed angles rather than continuously turning the card platform. The ULN2003 driver board connects the Arduino to the stepper motor and allows the Arduino to control the motor's stepping sequence.

The motor turns the physical platform, while the Arduino code controls direction, speed, and stopping behavior.

This motor choice also affected the experience of the project. The stepper motor does not move as silently or smoothly as a screen animation. Its slight mechanical rhythm, small pauses, and audible texture made the rotation feel more physical and object-like. Although this was partly a technical limitation, it also supported the ritual atmosphere of the device. The turntable felt like a small mechanism with weight and resistance, rather than a perfectly smooth digital effect.

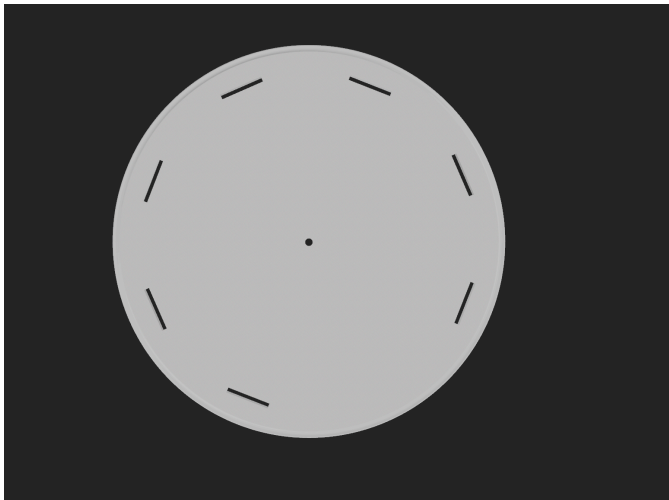
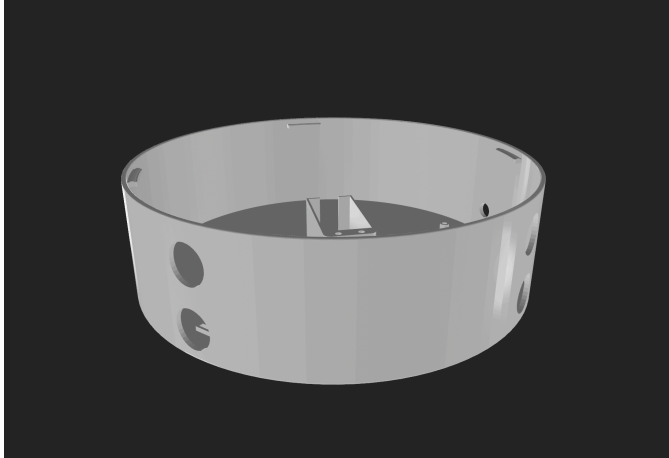
The overall input–process–output flow is:

Hand gesture → ultrasonic distance sensing → Arduino logic → stepper motor rotation → vibration feedback.

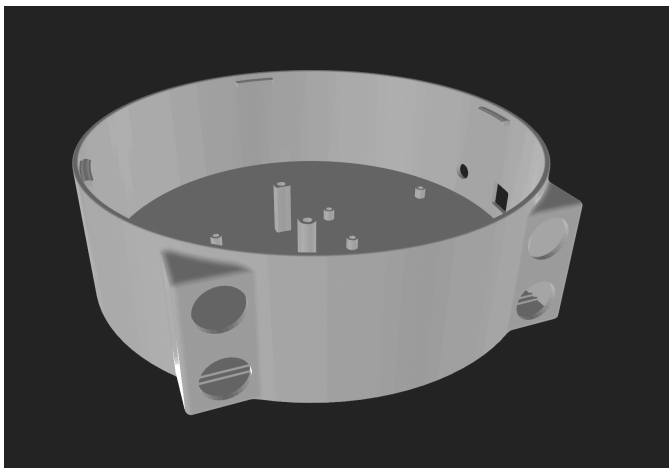
This system also revealed a common challenge in gesture-based interfaces. Catuncan and Ibrahim’s gesture-controlled robotic puzzle found that gesture control can support embodied interaction, but it can also introduce reliability and feedback challenges when users are unsure whether a gesture has been recognized . My prototype faced a similar issue. Because the sensing area is invisible, users sometimes placed their hands too far away or moved too quickly. This made sensor placement, threshold tuning, and physical guidance very important.

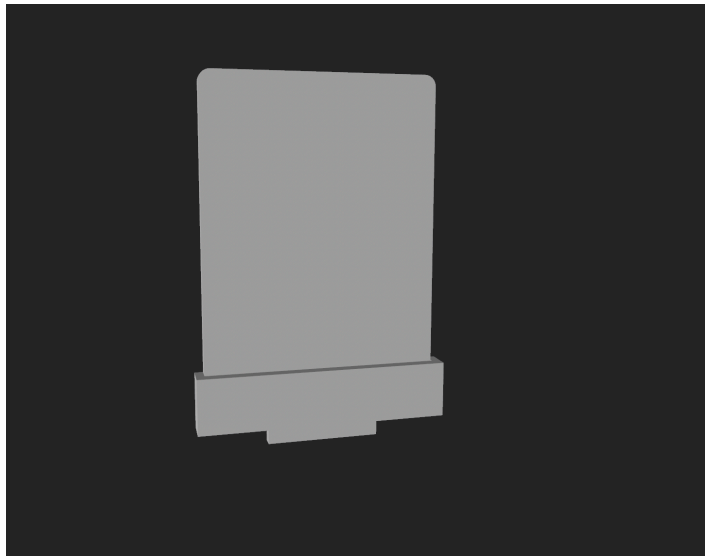
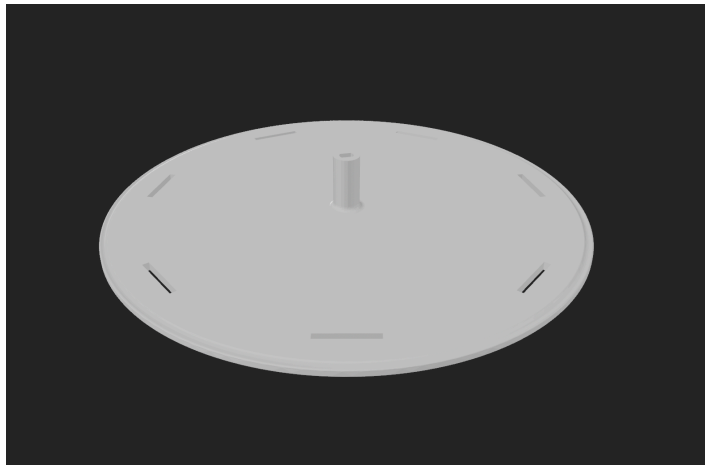
5. Photos of Prototypes, Wiring View, and TinkerCAD Circuit Diagram

Figure 1. prototype view



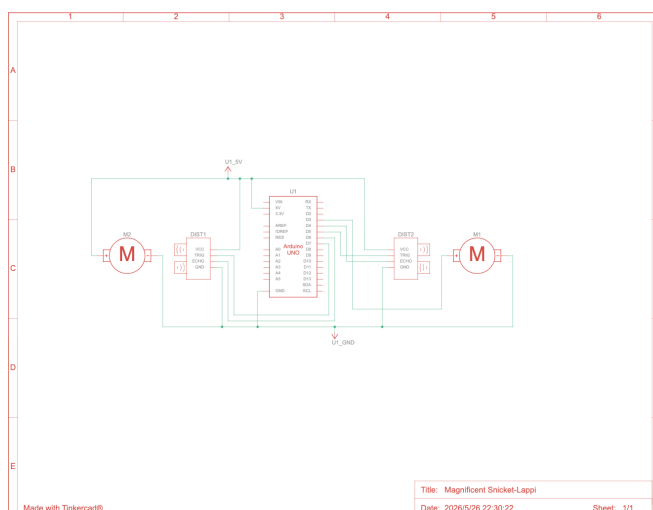
The first prototype used a servo motor, with sensors placed on both sides of a curved surface.

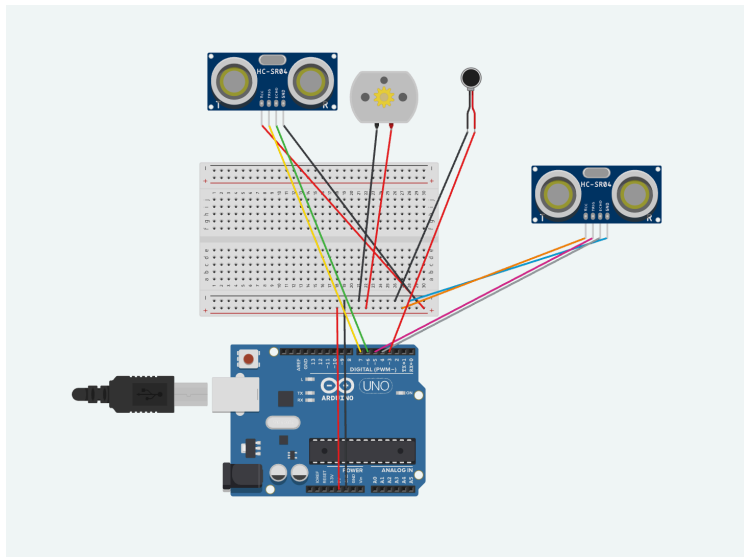




The final prototype

Figure 2. TinkerCAD circuit diagram





6. Reflection: Interaction with People

During prototype testing, people were generally interested in controlling the device and selecting cards through motion. The basic interaction was easy to understand: moving a hand near one side of the device would rotate the card carousel, and bringing both hands close would stop it. Users quickly understood that the system was not only displaying cards, but allowing them to physically participate in the selection process. This supported my original intention of making tarot selection feel more active, embodied, and time-based.

Several users spent time experimenting with different gestures, such as moving their hands quickly, hovering near the sensors, or using both hands to stop the turntable. This showed that the physical interaction itself was engaging, even before the card-reading content was fully developed. Some users were also curious about the content and meaning of the cards, which reminded me that tarot is not only a physical ritual, but also an interpretive system. Because of time limitations, this prototype focused more on the interaction mechanism—sensing, rotating, stopping, and vibrating—rather than detailed card interpretation. A future version should connect the selected card more directly to a meaningful reading experience.

However, one issue I did not fully expect was that the sensors were not sensitive or fluid enough to support the effortless magical control I imagined. When users knew the rule of the interaction but did not understand the sensing mechanism, they sometimes had difficulty triggering the system at the right moment. A card they wanted might have already rotated past before the sensor responded, or the vibration feedback might happen suddenly and surprise them. As a result, the interaction sometimes felt less natural than I intended. This helped me realize that a magical interface still needs very precise sensing and feedback. If the system is meant to feel like the user is controlling the object through invisible force, the sensing must be responsive enough that the user's gesture and the object's motion feel tightly connected.

This limitation also changed how I understood the carousel metaphor. The rotating cards successfully created anticipation, but the timing of control was not always smooth enough for users to feel fully connected to the moving cards. In a stronger version, the user should feel as if

they are gently touching or pushing the carousel through the air, rather than waiting for a sensor to detect them. The prototype therefore worked well as a conceptual and mechanical exploration, but the next step is to make the interaction feel more continuous, sensitive, and physically believable.

Some users also suggested that the project could be extended beyond tarot into other forms of card-based interaction, such as card games or party games. This feedback was useful because it showed that the rotating card mechanism has broader potential. The core interaction—gesturing to move a set of cards, stopping the motion, and receiving one selected card—could support many social or playful experiences. In this sense, the prototype is not only a tarot device, but also a platform for exploring actuated card selection.

Overall, the testing showed that the project’s strongest quality is the combination of physical motion and ritualized selection. People were interested not only in the final card, but also in the act of controlling, waiting, and stopping. This suggests that the carousel metaphor and tarot metaphor can work together: the rotating structure creates anticipation, while the card selection gives that anticipation a symbolic outcome.

7. Future Work and Implications

In future work, I would like to make the carousel and “wheel of fate” metaphor more complete. The current prototype places tarot cards on a rotating platform, but the next version could make

the cards feel more suspended, magical, and spatial. For example, I would like to explore a floating or semi-floating structure, where the cards appear to hover around the center rather than simply stand on a platform. This would strengthen the feeling that the cards are moving through an enchanted space, rather than being mechanically carried by a base.

Lighting would also be an important direction. Subtle light could follow the rotation, highlight the selected card, or respond to the user's hand movement. For example, when the user waves a hand near one sensor, the corresponding side of the turntable could glow softly; when both hands move close to stop the turntable, the selected card could be illuminated. This would make the moment of selection more theatrical and would help connect the user's gesture, the object's motion, and the final card reveal.

I would also like to make the non-contact control feel more physically believable, as if the user is truly pushing the card carousel through the air. The current version detects whether a hand is within a distance threshold, but it does not fully capture the dynamics of a real hand gesture. A more refined version could respond not only to presence, but also to gesture rhythm, repeated motion, and direction. For example, if the user waves in the same direction twice within a few seconds, the turntable could accelerate briefly and then gradually slow down, similar to the way a real carousel or wheel gains momentum after being pushed. This would make the interaction feel less like an on/off sensor trigger and more like a physical transfer of force.

I would also like to simulate more realistic motion qualities, such as acceleration, inertia, friction, and gradual deceleration. These details are important because the project is not only about making the turntable move, but about making the movement feel meaningful. If the turntable speeds up after repeated gestures and slowly comes to rest, the user may feel that their action has physically affected the object. This would strengthen the magical feeling of remote control while also making the carousel metaphor more complete. The goal is for the user to feel that the cards are not simply being driven by a motor, but are responding to the energy, timing, and rhythm of the hand gesture.

Another future direction is to integrate the actual meaning of the cards. In this prototype, the seven-card deck functioned mainly as a symbolic and visual structure. In a future version, each selected card could trigger a short interpretation, a sound, a light pattern, or a printed message. This would connect the physical act of stopping the carousel with the emotional and reflective part of tarot reading. The interaction would not end when the turntable stops; instead, the stopped card would open a moment of interpretation.

The broader implication of this project is that actuated everyday objects can turn selection into a meaningful experience. In many interfaces, selection is treated as a quick and efficient action: click, tap, choose. My project explores the opposite approach. It slows selection down and makes it physical. By combining carousel-like motion, non-contact gesture control, vibration feedback, and symbolic cards, the system shows how actuation can transform a simple choice into an embodied ritual.

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